density is as a stationary point of the Gibbs-Boltzmann entropy[4]

$$\mathrm{S} = -\mathrm{k_B} \int \rho \ln \rho \mathrm{d}\omega,$$

subject to the constraints

$$\int \rho(z)\mathrm{d}\omega = 1, \qquad \int H(z)\rho(z)\mathrm{d}\omega = \bar{\mathrm{E}},$$

where the first constraint amounts to a normalization to make ρ a probability density, whereas the second fixes the average energy of the system to a predefined value. It does not matter what this value is at this point.

This can be seen as a variational principle. Using the method of Lagrange multipliers, we seek stationary solutions of

$$V = \mathrm{S}/\mathrm{k_B} - \lambda \left[\int \rho(z)\mathrm{d}\omega - 1 \right] - \beta \left[\int H(z)\rho(z)\mathrm{d}\omega - \bar{\mathrm{E}} \right]$$

with respect to the choice of density ρ and coefficients λ and β. The resulting equations for the maximizing density are obtained by variational differentiation (see Sect. 2.2.1):

$$\frac{\delta V}{\delta \rho} = 0$$

$$\int \rho(z)\mathrm{d}\omega = 1$$

$$\int H(z)\rho(z)\mathrm{d}\omega = \bar{\mathrm{E}}.$$

The first equation reduces to

$$-(1 + \ln \rho) - \lambda - \beta H = 0$$

yielding

$$\rho = e^{-1-\lambda} e^{-\beta H}.$$

[4]This form of the entropy is more general than the "Boltzmann" form (6.1); it was introduced in Gibbs' treatise [147].

The normalization condition determines the parameter λ, whereas β and hence the temperature arises from the choice of average energy, i.e. from solving the equation

$$\frac{\int H(z)e^{-\beta H(z)}\mathrm{d}\omega}{\int e^{-\beta H(z)}\mathrm{d}\omega} = \bar{\mathrm{E}}$$

Alternatively, we may take the position that the temperature is specified and the mean energy is a direct consequence of this choice.

This defines the phase space density of the canonical ensemble which we refer to as the Gibbs-Boltzmann density:

$$\rho_\beta = Z_{\mathrm{can}}^{-1} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}. \tag{6.3}$$

where the parameter $\beta = 1/(\mathrm{k_B T})$. The quantity Z_{can} is a normalizing constant so that this is a probability density. The canonical ensemble is referred to as the NVT ensemble, due to the a constancy of the number of particles, volume, and temperature.

Now let us see the consequences of assuming that the states of a given system are canonically distributed. In order for the density in (6.3) to be normalizable, it is necessary that the quantity Z_{can} defined by

$$Z_{\mathrm{can}} = \int e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\mathrm{d}\omega$$

be finite, i.e., that the integral converge. Here the form of the system and the assumptions regarding boundaries come into play. If we have a standard molecular Hamiltonian $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ with $\mathrm{N_c} = \mathrm{N_d}$ degrees of freedom, then the integral separates

$$Z_{\mathrm{can}} = \int_{\mathcal{D}} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\mathrm{d}\omega = \int_{\mathbb{R}^{\mathrm{N_c}}} e^{-\beta \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2}\mathrm{d}\omega_p \times \int_{\mathcal{Q}} e^{-\beta U(\boldsymbol{q})}\mathrm{d}\omega_q,$$

where $\mathcal{Q}$ is the accessible position domain, $\mathrm{d}\omega_p = \mathrm{d}p_1\mathrm{d}p_2\ldots\mathrm{d}p_{\mathrm{N_c}}$ and $\mathrm{d}\omega_q = \mathrm{d}q_1\mathrm{d}q_2\ldots\mathrm{d}q_{\mathrm{N_c}}$. If U is smooth and the domain $\mathcal{Q}$ is finite, then clearly both integrals are well defined, so the normalization is sensible. When U has singularities, it is important that at these points $U \to +\infty$ instead of $-\infty$. Moreover, if no boundary conditions are employed to restrict the particles, we would need to introduce an additional restriction on the potential U in order to ensure that the configurational integral converges.

Typically one assumes that (i) U is smooth except on a low-dimensional set of singular points, (ii) U is bounded below (which is normally the case if a potential such as Lennard-Jones, steeply repulsive at short range, is incorporated between all pairs of atoms), and (iii) U grows sufficiently rapidly as $\boldsymbol{q} \to \infty$, in the case of an infinite domain.